

DataQuest Project Spec

Draft Document 2026

Abstract

This document represents the proposed project spec for the 2026 DataQuest project. The central theme of this year's project is implementing generative AI as tool to aid data scientists in the credit space of a bank

1. Student-Facing Project Brief

1.1. Project Title: DataQuest 2026 - Building Interpretable Credit Models

1.2. Background

You are part of the analytics team at a retail lending company. The company provides consumer loans and wants to improve how it evaluates loan applications.

You have been given a historical dataset containing past applicants, information available at the time of application, and whether each applicant defaulted within 12 months of taking the loan.

Your task is to investigate the data, identify risk patterns, and build an interpretable predictive model that helps the lender make better approval decisions.

The business does **not** want a non-linear solution (*e.g. Random Forest, Neural Network, etc.*) Instead, it wants a model that can be explained to risk managers, business stakeholders, and regulators. For that reason, you will work under a realistic modelling constraint: your final predictive model must be based on **logistic regression**.

1.3. Your Objective

There are two main tasks and an optional third bonus task.

1. Research & Building an interactive EDA tool to explore the data and identify meaningful risk patterns.
2. Improve a baseline logistic regression model through feature engineering and scorecard-style transformations.
3. Create a business decision dashboard that shows how your model would affect approvals, risk, and profitability.

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This is not only a modelling exercise. It is also a project in:

- Structured data exploration,
- Interpretable machine learning,
- Feature engineering,
- Business decision support,
- Using AI as a productivity companion.

1.4. *The Dataset*

The dataset provided is a simulated historical loan-book dataset. Note the data are entirely simulated and are not specific to any country or region.

1.5. *Important Note on Realism*

This dataset is intentionally messy and realistic. A superficial analysis is unlikely to be enough. Strong submissions will go beyond simple summary statistics.

2. Task 1 — Research & Interactive EDA Tool

2.1. *Goal*

The first goal is to research machine learning concepts that relate to credit modelling.

The second goal is to build an interactive **Exploratory Data Analysis** tool that helps a credit-risk analyst understand the dataset and identify useful patterns.

A static notebook alone is not enough for this task. You should create an app-style interface (for example, in Python using Streamlit or Shiny in R) that allows a user to interactively inspect variables and relationships.

2.2. *Your Research Section should include:*

- Brief paragraphs, equations, and plots that show the difference between *Generalised Linear Models* and *Non-Linear Models* specifically for classification.
- The balance between interpretability and complexity.
- A brief explanation of credit-modelling concepts such as Weights of Evidence Encoding and Information Value. Explain why these are useful in a credit space.
- A brief description on key metrics such as Accuracy, AUC, Gini, Precision, Recall and F_1 . Focusing on how these relate to credit modelling.
- Despite the fact that the data are simulated, research which feature(s) a regulator may disapprove of and why.

2.3. *Your EDA Tool should include:*

2.3.1. *Univariate explorer*

Allow the user to select a single feature and inspect:

- The feature distribution,
- How the feature distribution differs across the target.
- For credit-style analysis, include WoE (Weight of Evidence) and IV (Information Value) where appropriate.

2.3.2. *Bivariate explorer*

Allow the user to inspect relationships between two variables using:

- Scatterplots,
- Heatmaps,
- or similar techniques.

The goal here is to make interactions and subgroup effects visible.

2.3.3. *Additional*

- A data quality report
- Any other visualisation techniques you believe will be useful.

A strong submission will not only visualise the data, but will **extract insight** from it. Your tool should help you discover patterns that later inform your feature engineering in Task 2.

3. Task 2 — Improve the Baseline Logistic Regression

3.1. *Goal*

The goal is to build an improved **logistic regression** model that outperforms a baseline model implementing targeted and thoughtful feature engineering.

The following benchmark context is provided:

- **Baseline logistic regression AUC: 0.68**
 - an older model.
- **Reference LightGBM AUC: 0.82**
 - included only as a performance ceiling / benchmark.

You may **not** use LightGBM (or any other non-linear model) as your final model for this task.

Your challenge is to see how far you can push **logistic regression** through better transformations and domain-informed design.

3.2. *Expected approach*

Your EDA findings from Task 1 should directly shape your modelling strategy. In other words, this is not a “fit logistic regression and submit” task. It is a feature engineering task.

3.3. What we are looking for

We want to see:

- How you translate exploratory findings into features,
- Whether your transformations are sensible and interpretable,
- Whether they materially improve predictive performance.

A good submission will clearly explain:

- What was discovered in EDA,
- What adjustments were made.,
- What improvement these adjustments induced.

4. Bonus Task 3 — Business Value Dashboard

4.1. Goal

Models are only useful if they support decisions. In this task, the app should be extended to show how your model could be used in a lending workflow.

Your dashboard should help a business user understand,

- What approval strategy to use.
- How volume and risk change with policy.

You are free to innovate in this section.

4.2. Suggested Elements to Explore (Non-Exhaustive)

Your dashboard should encourage **policy-level thinking**, not just model performance.

4.2.1. Volume vs Risk Trade-off

Help the user reason about,

- How many customers are approved at different decision thresholds.
- How overall portfolio risk changes as approval volume increases.
- Whether pushing for higher volume disproportionately increases risk

4.2.2. Precision vs Recall — Business Interpretation

Surface the idea that common ML metrics have *business meaning*,

- What does **precision** mean in the context of approved loans?
- What does **recall** mean in terms of missed opportunities?
- Where is one error more *costly* than the other ?

5. Deliverables

Your final submission should include:

5.1. *Interactive App Code*

An application containing:

- EDA components.
- Model evaluation views.
- Business decision dashboard components (if done).

5.2. *Modelling summary*

A short written report or slide deck describing:

- Your research section.
- Your key EDA findings. Plots or screenshots from your EDA platform are required.
- Your cleaning approach.
- Your feature engineering choices.
- Your modeling methodology.
- Your final logistic regression model, in mathematical form i.e. the equation of the linear predictor η .
- Your performance relative to baseline,
- Your business recommendations (optional).

5.3. *Reproducible workflow*

Your project should be structured clearly enough that another analyst could understand and reproduce your steps. We would like you to attach your code so that results can be validated.

5.4. *AI usage reflection*

Document:

- which prompts you used,
- where AI helped,
- what suggestions you accepted,
- what you modified,
- and what you rejected.

This is not a penalty item. It is part of the learning goal. We want to see how you use AI as an analytical assistant rather than a substitute for judgment. We do not expect you to be familiar with all the concepts described above. How you tackle this uncertainty with AI is critical.

6. Constraints

- Your final predictive model for Task 2 must use **logistic regression**.
- Your work should emphasise **interpretability** and **clear reasoning**.
- Use the EDA to guide your modelling choices.
- Do not rely only on automated feature selection or trial-and-error tuning without explanation.

7. Optional Extensions / Bonus Ideas

If you want to go further, you may add:

- a scorecard-style points system based on model coefficients,
- explainability views (for example, feature contributions),
- a model comparison tab,
- scenario analysis under different loss assumptions.
- More advanced machine learning model results, to show “what’s possible”